

1759–60

30,000

THE NUMBER OF **PLANT SPECIES**
IN THE LIVING COLLECTIONS
AT **KEW GARDENS**

In 1760, the botanic gardens at Kew in London were enlarged to accommodate the many exotic plants brought back from distant lands. This is Kew's great Palm House of 1848.



H4 chronometer

The first practical device for calculating longitude at sea, Harrison's H4 chronometer was like a large pocket watch, 5 in (13 cm) across and weighing 3.2 lb (1.45 kg).

THE PROBLEM OF CALCULATING LONGITUDE AT SEA WAS

finally solved in 1759 with a highly accurate clock, or chronometer. Most people had assumed that such a clock would be large and complex. Between 1730 and 1759, English clockmaker **John Harrison** had built three chronometers, all very accurate but not accurate enough. Then Harrison realized the clock did not have to be big. In 1760, he built a **chronometer** the size of a pocket watch. Called H4, it worked astonishingly well, losing just 5.1 seconds in a two-month journey across the Atlantic in 1761.

As mariners sailed farther, they brought **exotic plants** to Europe from across the globe. These were planted in newly created botanical gardens, such as **Kew Gardens** in London, which was greatly enlarged in 1760 by Augusta of Saxe-Coburg, dowager Princess of Wales.

Mariners sailing through the Southern Ocean brought back tales of **giant icebergs**. Russian polymath **Mikhail Lomonosov** suggested that they must have formed on dry land, on a **continent as yet undiscovered**, which later proved to be **Antarctica**. He also suggested that some rocks were much older than others and that the **history**

“THE THEATER OF THE MIND COULD BE GENERATED BY THE MACHINERY OF THE BRAIN.”

Charles Bonnet, Swiss scientist, from *Essai Analytique sur les Facultés de L'âme* (Analytical Essay on the Powers of the Soul), 1760

of Earth's landscapes must be long and complex, not simply the result of a few brief catastrophes.

One year previously, Italian geologist **Giovanni Arduino** (1714–95) suggested that the **geological history of Earth** could be divided into four periods: Primitive, Secondary, Tertiary, and Volcanic or Quaternary.

In 1760, **Charles Bonnet** described what came to be known as **Charles Bonnet syndrome**, a condition in which people with poor eyesight are afflicted with hallucinations. He observed the symptoms in his grandfather and suggested that the visualizations that the mind sees are generated by the physical brain.

CHARLES BONNET (1720–93)

Born near Geneva, Charles Bonnet lived all his life in his hometown. His studies included research on parthenogenesis in insects (reproduction without sex) and the discovery that caterpillars breathe through pores. A naturalist as well as a philosopher, he also pioneered the idea that the mind is the product of the physical brain.



1759 Giovanni Arduino proposes dividing the geological history of Earth into four periods

1760 Kew Gardens, London, enlarged by Augusta of Saxe-Coburg

1760 English geologist John Michell suggests cause of earthquakes

1760 John Harrison completes H4 chronometer

1760 Mikhail Lomonosov suggests that Earth's landscape is created by long, slow natural processes